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Date: 5/1/26

ECE 3364 – Quiz #5

Spring 2026

Do not open this quiz until you are told to begin.

1. Print your name, and sign your name, at the top of this page.
2. This quiz is closed book, closed notes. You may use one 8.5" × 11" crib sheet, or its equivalent. You may use a calculator. You should not use a cell phone, tablet computer, or laptop computer as you work on this quiz.
3. Show all work on these pages, and do NOT use backside of each page. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit. You may separate the pages as you work.
4. Show all units in solutions, intermediate results, and figures. Units in the quiz will be included between square brackets.
5. Do not use red ink. Do not use red pencil.
6. You will have 30 minutes to work on this quiz.

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Room for extra work

A circuit has the transfer function, $T(\omega)$, given below. Assume that the straight-line approximations to the magnitude and phase Bode plots have been drawn. You do not need to draw these plots, but you may draw them if you wish to.

$$T(\omega) = \frac{-468 (j\omega)^3 \left(j\omega + 370 \left[\frac{\text{rad}}{\text{s}} \right] \right)^2}{\left(1 \left[\frac{\text{rad}}{\text{s}} \right] + j\omega 2.5 \right)^3 (j\omega)^2}$$

- a) (25 pts) Determine the values of the poles and zeroes of the transfer function $T(\omega)$. If there is more than one pole or zero at the same frequency, indicate how many poles or zeroes are at that frequency.

$$T\omega = \frac{-468 (j\omega)^3 \left(j\omega + 370 \left[\frac{\text{rad}}{\text{s}} \right] \right)^2}{(1 + 2.5j\omega)^3 (j\omega)^2} = \frac{-468 (j\omega) \left(j\omega + 370 \left[\frac{\text{rad}}{\text{s}} \right] \right)^2}{(1 + 2.5j\omega)^3}$$

$$\text{Zeroes}_1: \omega = 0 \quad \frac{(j\omega)^3}{(j\omega)^2}$$

$$\text{Poles: } \omega = \frac{1}{2.5} = 0.4 \frac{\text{rad}}{\text{s}}$$

triple poles

$$\text{Zeroes}_2: \omega = 370$$

double

- b) (25 pts) What is the slope of the straight-line approximation to the magnitude Bode plot at $\omega = 3 \text{ [rad/s]}$? How did you determine this slope? Explain your approach, briefly.

Starting from origin is $+20 \text{ dB/dec}$

Corner frequency: $0.4, 370$

at $\omega = 3$, only the triple pole at 0.4 rad/s has been passed

$$\therefore \text{Slope} = +20 + (-3 \cdot 20) = -40 \text{ dB/dec}$$

The triple pole at 0.4 rad/s adds -60 dB/dec , putting -40 dB/dec between 0.4 and $370 \frac{\text{rad}}{\text{s}}$.

- c) (25 pts) What is the slope of the straight-line approximation to the phase Bode plot at $\omega = 3$ [rad/s]? How did you determine this slope? Explain your approach, briefly.

For triple pole at 0.4 it is active at $\omega = 3$, other than the rest of the poles. so $(-3 \cdot 45^\circ)$, $45^\circ/\text{dec}$ being the phase between poles and zeros.

$$\therefore \text{Slope} = -135^\circ/\text{dec}$$

- d) (25 pts) Find the magnitude in [dB], and the phase in [degrees], of the transfer function at $\omega = 3$ [mrad/s].

$$\omega = 3 \text{ mrad/s} = 0.003 \text{ rad/s}$$

$$|j\omega + 370| \approx 370, \quad |1 + j2.5 \cdot 0.003| \approx 1$$

Magnitude:

$$|T| = \frac{468 \cdot 0.003 \cdot 370^2}{1^3} = 192,208$$

$$|T|_{\text{dB}} = 20 \log_{10}(192,208) = 105.7 \text{ dB}$$

$$\text{Phase } \angle T = -180^\circ + 90^\circ = -90^\circ$$

The -180° comes from the negative leading coefficient, and 90° from the zero at the origin.

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